

Math 534 H - Spring 2026 - Lecture 16

Aside: Let's revisit the wave equation and prove uniqueness using the energy for the wave EQ.
Let's assume we wts the uniqueness of solutions to

$$\begin{cases} \text{IVP(W)} & \left\{ \begin{array}{l} u_{tt} - c^2 u_{xx} = 0 \quad x \in \mathbb{R}, \quad t > 0 \\ u(x, 0) = \phi(x), \quad x \in \mathbb{R} \\ u_t(x, 0) = \psi(x), \quad x \in \mathbb{R} \end{array} \right. \end{cases}$$

Recall $E[u](t) = \frac{1}{2} \int_{\mathbb{R}} |u_t|^2 + |u_x|^2 dx$

And we proved $\frac{dE}{dt} = 0 \Rightarrow E[u](t) = E[u](0)$

Assume u_1 and u_2 satisfy $\text{EVP}(w)$ with the same IC. (ϕ, ψ) . Then let $w = u_1 - u_2$

we get
$$\begin{cases} w_{tt} - c^2 w_{xx} = 0 \\ w(x, 0) = 0 \\ w_t(x, 0) = 0 \end{cases} \rightarrow$$

By Energy conservation $0 \leq E[w](t) = E[w](0)$

$\Rightarrow 0 = E[w](t) = \frac{1}{2} \int_{\mathbb{R}} \overbrace{|w_t|^2}^{\geq 0} + \overbrace{|w_x|^2}^{\geq 0} dx \underbrace{= 0}_{=0}$
functions of (x, t) .

$\Rightarrow w_t(x, t) = 0$ and $w_x(x, t) = 0$

① $W_t(x, t) = 0 \Rightarrow W(x, t)$ is "constant" in t
(could be a function of x)

② $W_x(x, t) = 0 \Rightarrow W(x, t)$ is "constant" in x

① + ② $\Rightarrow W(x, t) = C$, $C = \underline{\text{constant}}$

But

$$0 = W(x, 0) = C$$

$\therefore C = 0$. Hence $W(x, t) = 0 \forall (x, t)$.

$$\Rightarrow u_1(x, t) = u_2(x, t) \quad \forall (x, t)$$

as desired.

We continue now with the Heat/Diffusion Equation
on the whole real line $\mathbb{R}$:

$$\text{Consider the IVP (H)} \quad \begin{cases} u_t - k u_{xx} = 0, & \underline{x \in \mathbb{R}}, t > 0 \\ u(x, 0) = \phi(x) & \underline{x \in \mathbb{R}} \end{cases}$$

$(\underline{k > 0})$

We want to find an expression for the solution $u(x, t)$ of the IVP (H) above on $\mathbb{R} \times [0, \infty)$.
 (x, t)

How do we do this?

Let's first understand the PROPERTIES of
solutions to $u_t - k u_{xx} = 0$, $k > 0$ $x \in \mathbb{R}$
 $t > 0$

i) If $u(x, t)$ is a solution then the translate $u(\underline{x-y}, t)$ is also a solution for any fixed y

ii) Any derivative of a solution u (for ex. u_x, u_t, u_{xx} , etc) is also a solution.

iii) Linear combination of solutions to $u_t - k u_{xx} = 0$ is again a solution of the heat/diffusion equation.

iv) An integral of solutions is again a solution.

Hence if $S(x, t)$ solves $S_t - k S_{xx} = 0$, then so is $S(x-y, t)$ - by i) - and so is

$$v(x, t) = \int_{-\infty}^{\infty} S(x-y, t) f(y) dy \text{ for any}$$

function $g(y)$ - as long as the integral converges in a suitable sense.

• Although iv) might seem mysterious one can indeed prove this as a consequence of (iii).

v) If $u(x, t)$ is a solution of $u_t - k u_{xx} = 0$ then the dilated function $u(\sqrt{\lambda} x, \lambda t)$ for any $\lambda > 0$ fixed is also a solution.

This follows from chain rule:

Let $v(x, t) = u(\sqrt{\lambda} x, \lambda t)$, then $v_t = \lambda u_t$ and $v_{xx} = \lambda u_{xx}$, so $v_t - k v_{xx} = \lambda (u_t - k u_{xx}) = 0$ ($\lambda > 0$)

Now using the properties above we would like to find a particular solution of

$$(k > 0) \quad u_t - k u_{xx} = 0, \quad x \in \mathbb{R}, \quad t > 0.$$

The particular solution we seek will have as initial conditions the Heaviside function

$$H(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$$

NOTE

(*)

$$H(\lambda x) = H(x)$$

Let's denote by $Q(x, t)$ this particular solution

$$\text{Then } Q(x, 0) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}.$$

We want a representation formula for $Q(x, t)$.

If we succeed in doing so, then we can use it in conjunction with property iv) to find a representation formula for the general IVP (H) above.

So, we want to solve explicitly ($k \geq 0$)

$$(H) \begin{cases} \textcircled{1} \longrightarrow Q_t - k Q_{xx} = 0, x \in \mathbb{R}, t \geq 0 \\ \textcircled{2} \longrightarrow Q(x, 0) = H(x), x \in \mathbb{R} \end{cases}$$

Note: From the dilation invariance (v) we know

that $Q(\sqrt{\lambda} x, \lambda t)$ also solves $\textcircled{1}$ with

$\textcircled{2}$ being $Q(\sqrt{\lambda} x, 0) = H(\sqrt{\lambda} x) = H(x)$

by $\textcircled{*}$

Hence both $Q(\sqrt{\lambda}x, \lambda t)$ and $Q(x, t)$
solve the SAME IVP (1) + (2) (SAME IC!)

But since solutions are unique (for same IC)

we must have that $Q(\sqrt{\lambda}x, \lambda t) = Q(x, t)$
for all $x \in \mathbb{R}$, $t > 0$ and $\lambda > 0$.

This tells us that Q itself must be
invariant under the dilation $(x, t) \rightarrow (\sqrt{\lambda}x, \lambda t)$

The only way this can happen is if Q
depends only on the ratio $\frac{x}{\sqrt{t}}$: that is

if $\boxed{Q(x, t) = g\left(\frac{x}{\sqrt{t}}\right)}$ (self-similar variable)
for some function g of a SINGLE VARIABLE

[You'll prove this in the next HWK set] We need to find such g to determine $Q(x, t)$.

• Now just for convenience in future calculations we pass from g to g where $g(z) = g(\sqrt{4k} z)$ so that:

•
$$Q(x, t) = g\left(\frac{x}{\sqrt{t}}\right) = g\left(\frac{x}{\sqrt{4kt}}\right) = g(p)$$

•
$$\boxed{p := \frac{x}{\sqrt{4kt}}}$$

$\sqrt{t} \cdot \sqrt{4k}$

Using this we can now convert the PDE for Q into an ODE for g :

$$Q_t = \frac{dg}{dp} \cdot \frac{dp}{dt} = -\frac{1}{2t} \cdot \frac{x}{\sqrt{4kt}} \cdot g'(p)$$

$$Q_x = \frac{dg}{dp} \cdot \frac{dp}{dx} = \frac{1}{\sqrt{4kt}} \cdot g'(p)$$

$$Q_{xx} = \frac{dQ_x}{dp} \cdot \frac{dp}{dx} = \frac{1}{4kt} g''(p) \quad \underline{\text{Hence}}$$

$$0 = Q_t - k Q_{xx} = \frac{1}{t} \left[-\frac{1}{2} p g'(p) - \frac{1}{4} g''(p) \right]$$

So we need to solve:

$$(ODE) : \quad \boxed{g'' + 2p g' = 0} \quad (g = g(p))$$

Using the integrating factor $e^{\int 2p \, dp} = e^{p^2}$
we get after some calculations (CLASS HWK!
you do)

$$Q(x,t) = g(p) = c_1 \int e^{-p^2} dp + c_2$$

We now have an explicit formula for $Q(x,t)$ by
recalling that $p = \frac{x}{\sqrt{4kt}}$:

$$Q(x,t) := c_1 \int_0^{\frac{x}{\sqrt{4kt}}} e^{-p^2} dp + c_2, \quad \underline{t > 0}$$

(notice we chose a particular antiderivative which we are free to do
due to the presence of the arbitrary constants c_1, c_2)

Now to check the IC we need to take $t \rightarrow 0^+$

Recall we have

$$\begin{aligned} \text{if } x \geq 0, \quad 1 &= \lim_{t \rightarrow 0^+} Q(x, t) = c_1 \int_0^{\infty} e^{-p^2} dp + c_2 \\ &= c_1 \frac{\sqrt{\pi}}{2} + c_2 \end{aligned}$$

$$\begin{aligned} \text{if } x < 0, \quad 0 &= \lim_{t \rightarrow 0^+} Q(x, t) = c_1 \int_0^{-\infty} e^{-p^2} dp + c_2 \\ &= -c_1 \frac{\sqrt{\pi}}{2} + c_2 \end{aligned}$$

Hence :

$$\begin{cases} 1 = c_1 \frac{\sqrt{\pi}}{2} + c_2 \\ 0 = -c_1 \frac{\sqrt{\pi}}{2} + c_2 \end{cases} \Rightarrow \begin{cases} c_1 = \frac{1}{\sqrt{\pi}} \\ c_2 = \frac{1}{2} \end{cases}$$

So all in all we have that the unique solution

$Q(x, t)$ to the **IVP (†) ①+②** above is

$$Q(x, t) = \frac{1}{2} + \frac{1}{\sqrt{\pi}} \int_0^{\frac{x}{\sqrt{4kt}}} e^{-p^2} dp \quad \text{for } \underline{t \geq 0}$$

Next: Want to solve the general IVP (†)

using this $Q(x, t)$ and the properties we saw before.